O-E-661D
April 17, 1974
SUPERSEDING
Fed. Spec. O-E-661C
April 5, 1965

FEDERAL SPECIFICATION

ERADICATOR, INK

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers requirements for ink eradicators for removing writing, copying, and record inks from paper and an eradicator for removing drawing ink from tracing cloth (see 6.1).

1.2 Classification.

1.2.1 Ink eradicator.

1.2.1.1 Grade, types, classes, and formulations. The ink eradicator shall be of one grade and shall be furnished in the following types, classes, and formulations, as specified (see 6.2):

Type I - One-solution unit.

Class 1 - For use on paper (as specified herein).
Class 2 - For use on tracing cloth (as specified herein).

Formulation a - (See 3.3 and 3.5).
Formulation b - (See 3.3 and 3.5).

Type II - Two-solution unit.

No. 1 Solution.
No. 2 Solution.

1.2.2 Containers of eradicator (sizes of bottles). (See 5.1.1.1.1 and 5.1.1.1.2).

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, or request for proposal form a part of this specification to the extent specified herein:

Federal Specifications:

TT-I-521 - Ink, Copying and Record.
TT-I-528 - Ink, Drawing, Waterproof, Black.
TT-I-563 - Ink, Writing, for Fountain and Dip Pens.
UU-P-121 - Paper, Bond and Writing, White and Colored.
UU-P-561 - Paper, Tracing.
CCC-C-531 - Cloth, Tracing.
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-665 - Boxes, Paperboard, Metal Edged and Components.
PPP-B-676 - Boxes, Set-Up.

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).
Fed. Std. No. 191 - Textile Test Methods.

FSC 7510

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(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA .

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The documents listed below form a part of this specification to the extent specified herein. Unless otherwise specified, the issue in effect on date of invitation for bids shall apply.

Code of Federal Regulations:

Part 191, Chapter I, Title 21 - Food and Drug Administration Regulations under the Federal Hazardous Substances Labeling Act.

(Copies and price of this document may be obtained from the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

National Motor Freight Traffic Association, Incorporated, Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N. W., Washington, D. C. 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Materials. Materials used in the specified solutions shall be such that the eradicator meets the requirements of this specification.

3.2 Sediment. There shall be not more than a trace of sediment (as received) in type I ink eradicator and in the No. 1 solution of type II eradicator.

3.3 pH. The pH of type I, class 1 ink eradicator and the No. 2 solution of type II eradicator shall be not less than 9.0 and not more than 10.5. The pH of type I, class 2, formulations a and b ink eradicator shall be not less than 7.0 and not more than 10.5.

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3.4 Odor. There shall be no pronounced objectionable odor.

3.5 Performance.

3.5.1 Type I, one-solution unit.

3.5.1.1 Ink eradicating properties on white bond and tracing papers, and tracing cloth. The eradicator shall easily and completely remove the ink, both before and after 7 days storage, and the sample where treated with one-solution eradicator shall show not more than a slight change from that portion of the paper or cloth, as applicable, not treated, when tested as specified in 4.4.1.

3.5.2 Type II, two-solution unit.

3.5.2.1 Ink eradicating properties on white bond paper. The eradicator shall easily and completely remove the ink, both before and after 7 days storage, and the paper where treated with the two-solution eradicator shall show not more than a slight change from that portion of the paper not treated, when tested as specified in 4.4.2.

3.5.2.2 Feathering. There shall be not more than a slight feathering or blurring of the lines in the treated section of the paper, when the two-solution eradicator is tested as described in 4.4.3

3.6 Identification of product marking or labeling. Bottles of eradicator shall be marked or labeled in accordance with normal commercial practice.

3.7 Instructions for use of product. Instructions for use of the eradicator shall be furnished in accordance with normal commercial practice.

3.8 Workmanship. Workmanship shall be in accordance with normal commercial practice. Labels, when used, shall be neatly and securely affixed and clearly printed, and bottles shall show no evidence of spilling or leaking of eradicator solution.

4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements (see 6.2).

4.2 Sampling and inspection for acceptance. The sampling and inspection shall be performed in accordance with the provisions of MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Lot. For purposes of sampling, the manufacturer shall submit the delivery in sampling lots, each of which is composed of products produced under essentially the same conditions.

4.2.1.1 Batch process. If the product is manufactured by the batch process, each batch shall constitute a lot.

4.2.1.2 Continuous process. If the product is manufactured by a continuous process, the lot shall consist of a product made from a single batch of raw materials or during a single production shift.

4.3 Inspection.

4.3.1 Inspection of materials and components. The supplier is responsible in accordance with 4.1 for insuring that components and materials used were manufactured, tested, and inspected in accordance with the specified requirements of referenced subsidiary specifications and standards and in accordance with this specification, as applicable.

4.3.2 Inspection of end item and preparation for delivery.

4.3.2.1 Examination of the end item. The end item shall be examined in accordance with the classification of defects listed for each of the applicable examinations, as indicated below.

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4.3.2.1.1 Examination of (type I) filled bottles for defects in appearance, design, workmanship, and marking or labeling. Examination shall be made as indicated in table I. The sample unit for this examination shall be one filled bottle of ink eradicator. The inspection level and acceptable quality level (AQL) shall be as specified in 4.3.2.1.4 with the lot size expressed in units of bottles.

4.3.2.1.2 Examination of (type II) filled bottles for defects in appearance, design, workmanship, and marking or labeling. Examination shall be made as indicated in table I. The sample unit for this examination shall be one set (two bottles) of filled containers of ink eradicator. The inspection and acceptable quality levels shall be as specified in 4.3.2.1.4 with the lot size expressed in units of bottles.

TABLE I. Examination of end item(s)

Examine	Defects
Appearance of ink eradicator.	More than a trace of sediment (see 3.2).
Bottles (individual containers).	Bottle material and closure not as specified or not commonly used for ink eradicator. Size not as specified.
Fill of individual bottle(s)(except average contents).	Empty bottle, or overfilling to extent that bottle may burst or leak in hot weather due to lack of liquid expansion space. Stray spilling of eradicator or leakage on outside of bottle(s).
Identification marking or labeling on bottle.	Not as specified.
Workmanship.	Not as specified.

4.3.2.1.3 Examination of preparation for delivery. An examination shall be made to determine compliance with the packaging, packing, and marking requirements of section 5 and contract or order. Defects shall be scored in accordance with table II. For examination of contents, the sample unit shall be one shipping container fully prepared for delivery, selected just prior to the closing operations. Defects of closure specified shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end-item inspection lot. The inspection level shall be S-4 with an acceptable quality level (AQL) of 4.0 defects per 100 units.

TABLE II. Examination of preparation for delivery

Examine	Defects
Markings- (unit and intermediate packages and exterior shipping containers, as applicable).	Incorrect; omitted; illegible; improper size, location, sequence, or method of application; omission "THIS SIDE UP" and other warnings.
Materials- (unit and intermediate packages and exterior shipping containers, as applicable).	Wrong type, class, or other category of required specification.
	Material missing, damaged, or defective affecting serviceability.
Workmanship	Wrong or careless means in placement and use of labeling and packaging and packing materials such as loose or partially attached labels; flaps of containers incompletely closed; loose or overtight straps, dangerous strap ends, protruding nails, and smeared adhesive.
Contents	Quantities of unit and intermediate packages not as specified. Wrong arrangement; bottle on side or upside down.
Closure of shipping containers.	Not as specified.
Gross weight of shipping containers	In excess of specified weight. In excess of box limitation shown on boxmaker's certificate.

4.3.2.1.4 Inspection levels and AQL's. The inspection levels and acceptable quality levels for examination of the end item shall be as follows:

Reference paragraph	Inspection Level	AQL
4.3.2.1.1	S-4	2.5
4.3.2.1.2	S-4 (See 4.3.2.1.3)	2.5

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4.3.3 Testing of ink eradicator. Each sample unit shall be taken at random from each lot or batch of ink eradicator and shall be tested in accordance with 4.4. Each sample unit shall meet all test requirements. One sample unit shall consist of two bottles of one-unit solution or two bottles each of the two-solutions, as applicable. (see 4.4.5).

4.4 Tests. During the following tests, sample eradicator shall be inspected for conformance with 3.4.

4.4.1 Ink eradicating properties (type I, one-solution unit). Eradicator shall be tested with materials specified in 4.4.1.1 in accordance with procedures described in 4.4.1.3.

4.4.1.1 Materials for tests (of type I eradicator).

4.4.1.1.1 Inks. Inks used in testing type I, class 1 eradicator shall conform to TT-I-521 and TT-I-563, class 1. Drawing ink for testing type I, class 2, formulations a and b sample eradicator shall conform to TT-I-528.

4.4.1.1.2 Paper, bond and tracing. Bond paper shall conform to UU-P-121, type III, white, 32-pound. Tracing paper shall conform to UU-P-561, type I, III, or IV, class 1 or 2, white.

4.4.1.1.3 Tracing cloth. Tracing cloth for testing type I, class 2, formulation "a" eradicator shall conform to type I or II of CCC-C-531, white, cotton; for testing type I, class 2, formulation "b" eradicator, type III of CCC-C-531, white cotton, shall be used.

4.4.1.2 Preparation of test areas (for procedures specified in 4.4.1.3)

4.4.1.2.1 Type I, class 1 eradicator. With pen and applicable ink (see 4.4.1.1.1), draw several cross section areas on bond or tracing paper, as applicable, making parallel lines 1/8 inch apart.

4.4.1.3 Procedures.

4.4.1.3.1 Type I, class 1 eradicator. Near the center of two or more areas of cross section, apply sample eradicator with glass or plastic rod,or in accordance with the manufacturer's instructions, if different, and blot after 5 seconds; if necessary, repeat once only. Store the test material in a well-ventilated room at 20° to 30°C. for a period of 7 days. After the specified period, again apply the eradicator to other areas of cross section as previously directed. Note the ease of ink eradication in comparison with test made previous to the 7-day storage period. Examine test paper, as applicable, upon completion of each test for conformance with the requirements of 3.5.1.

4.4.1.3.2 Type I, class 2 eradicator. Slightly moisten a clean, soft cloth with sample eradicator and rub moistened cloth over inked lines to be eradicated, near the center of two or more areas of cross section; repeat once if necessary, using clean part of the moistened cloth. (The manufacturer's instructions, if different, may be used). Store the test material in a well-ventilated room at 20° to 30°C for a period of 7 days. After the specified period, again apply the eradicator to other areas of cross section as previously directed. Note the ease of ink eradication in comparison with test made previous to the 7-day storage period. Examine test material upon completion of each test for conformance of eradicator with the requirements of 3.5.1.

4.4.2 Ink eradicating properties (type II, two-solution unit). Eradicator shall be tested with materials specified in 4.4.2.1 in accordance with procedure described in 4.4.2.3.

4.4.2.1 Materials for test (of type II eradicator).

4.4.2.1.1 Inks. Inks used in testing type II eradicator shall be as specified in 4.4.1.1.1 for testing type I, class 1 eradicator.

4.4.2.1.2 Paper, bond. Bond paper shall be as specified in 4.4.1.1.2.

4.4.2.2 Preparation of test areas (for procedure specified in 4.4.2.3.).

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4.4.2.3 Procedure. Near the center of two or more areas of cross section, apply solution No. 1 of type II eradicator with glass or plastic rod, and blot after 5 seconds; then apply solution No. 2 of the two-solution unit, and after the ink has disappeared, apply solution No. 1 again; mix and blot. (The manufacturer's instructions, if different, may be used). Store the test paper in a well-ventilated room at 20° to 30°C. for 7 days. After the specified period, again apply the eradicator, as directed, to other areas of cross section lines. Note the ease of ink eradication in comparison with test made to the 7-day storage period. Examine test paper upon completion of each test for conformance with the requirements of 3.5.2.

4.4.3 Feathering (type II only). Treat small areas of paper (see 4.4.2.1.2) with eradicator, using solutions No. 1 and 2 as directed, and allow the paper to become thoroughly dry. With pen and ink (see 4.4.2.1.1) draw lines through the treated areas and extending beyond on the untreated paper. Compare the appearance of the lines in the treated areas with those in the untreated areas.

4.4.4 pH. The pH of sample eradicator shall be determined, using electrometric measurement with a glass electrode, in accordance with applicable requirements of method 2811 of Fed. Std. No. 191.

4.4.5 Acceptance criteria. If any sample unit of ink eradicator selected in accordance with 4.3.3 fails to meet any of the tests of 4.4 through 4.4.4, the lot or batch represented by the defective sample unit shall be considered as not complying with the requirements of this specification, and the representative lot or batch shall be rejected.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit packages.

5.1.1.1.1 Type I, one-solution unit.

5.1.1.1.1.1 Class 1. The ink eradicator shall be furnished in one- and in one-half-ounce quantities, as specified (see 6.2).

(a) One-ounce quantity. The eradicator shall be furnished in glass bottles. The bottles shall be closed with a suitable rubber stopper, or composition screw cap, to which is fastened a glass rod applicator, or closed with a plastic cap and plastic applicator of one-piece construction or a plastic cap with plastic applicator fastened to the cap. The contents of the bottle shall be protected against light by use of amber colored glass or other effective means. The filled bottles shall be individually packaged in close-fitting boxes conforming to PPP-B-566, style II; PPP-B-665, style optional; or PPP-B-676, type optional. The boxes shall be closed in accordance with the applicable specification or appendix thereto.

(b) One-half-ounce quantity. Unless otherwise specified (see 6.2), type I, class 1 eradicator shall be furnished in polyethylene bottles. The bottles shall be of the flexible squeeze type with a plastic applicator tip which will dispense limited quantities of solution without flooding the writing. The applicator tip shall be made of polyethylene or urea formaldehyde plastic. A urea formaldehyde plastic cap shall be provided which screws to the bottle and effectively seals the applicator tip when in place. When specified (see 6.2), one-half-ounce quantities of ink eradicator shall be furnished in glass bottles with applicators and caps as specified for one-ounce bottles in 5.1.1.1.1.1 (a).

5.1.1.1.1.2 Class 2. The ink eradicator shall be furnished in one- and in four-ounce quantities, as specified (see 6.2). Unit packaging of the eradicator shall be as specified in 5.1.1.1.1.1 (a), except that four-ounce bottles shall have a suitable rubber stopper or metal screw cap conforming to normal commercial practice for this type of closure, or may be furnished with cap and applicator as specified for the one-ounce bottles.

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5.1.1.1.2 Type II, two-solution unit. Type II, two solution ink eradicator shall be furnished in 3/4 ounce quantities. Each solution (No. 1 and No. 2) shall be packaged separately in glass bottles. The bottles shall be closed with rubber stoppers or composition screw caps to which are attached glass rod applicators; or closed with a plastic cap and plastic applicator of one-piece construction or a plastic cap with plastic applicator fastened to the cap. The contents of the No. 2 bottles shall be protected against light by use of amber-colored glass or other effective means. The filled bottles shall be packaged in sets, one bottle of each solution (No. 1 and No.2) per set, in close fitting boxes conforming to PPP-B-566, style I; PPP-B-665, style optional; or PPP-B-676, type optional. A paperboard separator shall be placed between the bottles in the box. The separator shall be not less than two-thirds of the height of the bottles and shall fit snug against the sides of the box.

5.1.1.2 Intermediate packages. Twelve unit packages of either type I or type II eradicator, as applicable, packaged as specified in 5.1.1.1, shall be intermediate-packaged in close-fitting boxes conforming to PPP-B-566, style II; PPP-B-665, style optional; or PPP-B-676, type optional. The boxes shall be closed in accordance with the applicable box specification or appendix thereto.

5.1.2 Level B. Level B unit and intermediate packaging shall be as specified for level A (see 5.1.1 through 5.1.1.2).

5.1.3 Level C. The ink eradicators shall be packed in accordance with normal commercial practice.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Unless otherwise specified (see 6.2), 12 intermediate packages (144 units), packaged as specified in 5.1.1.2, shall be packed in snug-fitting boxes conforming to PPP-B-636, class-weather-resistant, V3c, style RSC. Each box shall be closed, waterproof sealed with tape, and reinforced with bands of filament tape in accordance with the appendix to the box specification.

5.2.2 Level B. Unless otherwise specified (see 6.2), 12 intermediate packages (144 units), packaged as specified in 5.1.1.2, shall be packed in snug-fitting boxes conforming to PPP-B-636, class-domestic, style RSC. The boxes shall be closed and sealed in accordance with the appendix to the box specification.

5.2.3 Level C. The ink eradicators shall be packed in accordance with normal commercial practice and in a manner to insure carrier acceptance and safe delivery at destination in containers complying with the rules and regulations applicable to the mode of transportation.

5.3 Marking or labeling. (see 3.6).

5.3.1 Civil agencies. In addition to any special marking required in the invitation for bids, contract, or order, intermediate boxes and exterior shipping containers shall be marked or labeled, as applicable, in accordance with Fed. Std. No. 123, including specification classification such as type, class, and formulation of eradicator, size of bottle, etc. Containers, as applicable, shall be marked or labeled in accordance with the Food and Drug Administration Regulations under the Federal Hazardous Substances Labeling Act. Shipping containers shall be appropriately marked "THIS SIDE UP."

5.3.2 Military agencies. In addition to any special marking required in the invitation for bids, contract, or order, intermediate packages and exterior shipping containers shall be marked or labeled, as applicable, in accordance with MIL-STD-129. Containers as applicable shall be marked or labeled in accordance with the Food and Drug Administration Regulations under the Federal Hazardous Substances Labeling Act. Shipping containers shall be appropriately marked "THIS SIDE UP".

5.4 Standard pack (civil agencies). The standard pack requirements for the commodity covered by this specification shall be as specified in 5.1.2, 5.2.2, and 5.3.1 (see 6.2.1).

6. NOTES

6.1 Intended use.

6.1.1 Type I, class 1, and type II ink eradicators are intended for use on white bond and tracing papers and other papers suitable for writing on with writing, copying and record inks.

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6.1.2 Type I, class 2. Type I, class 2, formulation "a" eradicator is intended for use on white tracing cloth such as types I and II of Federal Specification CCC-C-531; formulation "b", for use on tracing cloth as specified for type III in the cited specification. (Class 2 eradicators are not recommended for use on materials such as wool or silk).

6.2 Ordering data. Purchasers should select the preferred options permitted herein (see 5.1.1.1.1.1 (b), 5.2.1, and 5.2.3), and include the following in procurement documents:

(a) Title, number, and date of this specification.
(b) Type of eradicator, class, and formulation of class 2 required (see 1.2.1.1).
(c) Size of type I eradicator required (see 5.1.1.1.1.1 and 5.1.1.1.1.2).
(d) Data, as required; inspection responsibility if other than as specified (see 4.1).
(e) Levels of packaging and packing and any special marking for shipment required (see 5.1, 5.2, and 5.3).

6.3 Federal specifications do not include all types, classes, grades, sizes, etc., of commodities indicated by the titles of the specifications, or all those which are commercially available, but are intended to cover the types, etc., which are suitable for Federal Government requirements.

DOD Coordination has been waived.

User interest:

Army - GL

Preparing activity:

GSA - FSS

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